

ADERSH MARUVATTU

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Research Scholar, Robotics and Automation  
Co-Founder and CTO, Ceres FieldCheck

## PERSONAL STATEMENT

Robotics researcher and agricultural technology innovator with expertise in autonomous systems and machine learning. As a PhD candidate at IIT Kanpur, I'm developing advanced ML-based navigation systems for mobile manipulators traversing complex terrain. As CTO and co-founder of Ceres FieldCheck, I lead the technical development of affordable agricultural robots that combine AI-powered crop health inspection with precision weed removal technology for small-scale farmers.

My mission is to bridge the gap between cutting-edge robotics research and practical implementation, creating autonomous systems that solve real-world challenges. By integrating machine learning with robust control systems, I'm working to make agricultural robotics more accessible, sustainable, and impactful for farmers worldwide.

## COLLEGE EDUCATION

**IIT KANPUR**

2020 - Present

Kanpur, UP

PhD in Mechanical Engineering

- Research on robotic path planning on rough terrain
- Focus on mobile manipulators
- Rough terrain path planning using Machine learning and Model Predictive Control

**NIT CALICUT**

2013 - 2017

Calicut, KL

B.Tech in Mechanical Engineering

- Graduated with First Class
- Team Captain - SAE India Collegiate Club

## KEY COURSEWORK

**COURSES MASTERED**

- Intelligent Control Systems
- Introduction to Robotics
- Robot motion planning
- Robot manipulator dynamics and control
- Optimization methods in engineering
- Advanced Reinforcement learning for Robotics

**PROJECTS**

- B.Tech Project : The design used real-time monitoring of tool-work distance to ensure proper machining without tool buckling. The setup was used to drill a microhole in silicone borate.

## TECHNICAL SKILLS

MATLAB

LaTeX

SOLIDWORKS

COPPELIASIM

PYTHON

CURA (3D PRINTING)

ROS2

ARDUINO

## ASSOCIATED ORGANISATIONS

### ACADEMIC



### CORPORATE



### EVENTS



### NGO



## CORPORATE EXPERIENCE

**MARUTI SUZUKI INDIA LIMITED**

Jul 2017 - Nov 2019

Gurugram, NCR & Thiruvananthapuram, KL

Assistant Manager

- Increasing the efficiency of pick to light systems at Assembly line in Gurgaon plant
- Managing the worst affected workshop of the Kerala floods and meeting all delivery commitments
- Best Graduate Trainee for the year 2017
- Best Employee of the quarter 2018

## ROLES & RESPONSIBILITIES

**MATHEMATICS COORDINATOR**

Present

IITK, UP

Ranjit Singh Rozi Shiksha Kendra (Non-profit NGO under Unnath Bharath Abhiyan)

- Math coordinator for NGO takes and organizes 304 classes monthly for nearby public schools

**TEAM CAPTAIN - NITC**

2016

Pithampur, MP

BAJA SAE India

- Biggest Student off-roading event in India

**EXECUTIVE MEMBER**

2016

Calicut, KL

Mechanical Engineering Association, NITC

- Association for workshops and events related to Mechanical Engineering

## ACADEMIC PUBLICATIONS

**Maruvattu, Adersh, and Ashish Dutta. "End Effector Path Tracking by a 13-DOF Mobile Manipulator System Using ANN-Based Model Predictive Control." 2024 13th International Workshop on Robot Motion and Control (RoMoCo). IEEE, 2024.**

**Energy aware trajectory optimization using DRL in humanoid robot through via-point learning, James Sorokhaibam, Adersh Maruvattu, Ashish Dutta, ASME JMR**

**Under Review - Slip-Adaptive MPC for End-Effector Tracking of a Mobile Manipulator on Deformable Terrain, Adersh Maruvattu, Ashish Dutta, ASME JMR**

## AWARDS & RECOGNITIONS

**WINNER - DWIH INNOVATORS CONNECT**

2023

Bangalore and Stuttgart

For the best startup idea which was presented at the pitch competition

**BEST EMPLOYEE - MARUTI SUZUKI**

2018-19

Thiruvananthapuram, KL

For the work done at the biggest flood affected workshop in the Kerala floods of 2018